

Conceptual Scenario Supporting Extension of C2SIM to Autonomous Systems



NATO M&S CoE



Lt.Col. Marco BIAGINI
Maj. Fabio CORONA

Dr. Ulrich SHADE
Magdalena WOLSKI

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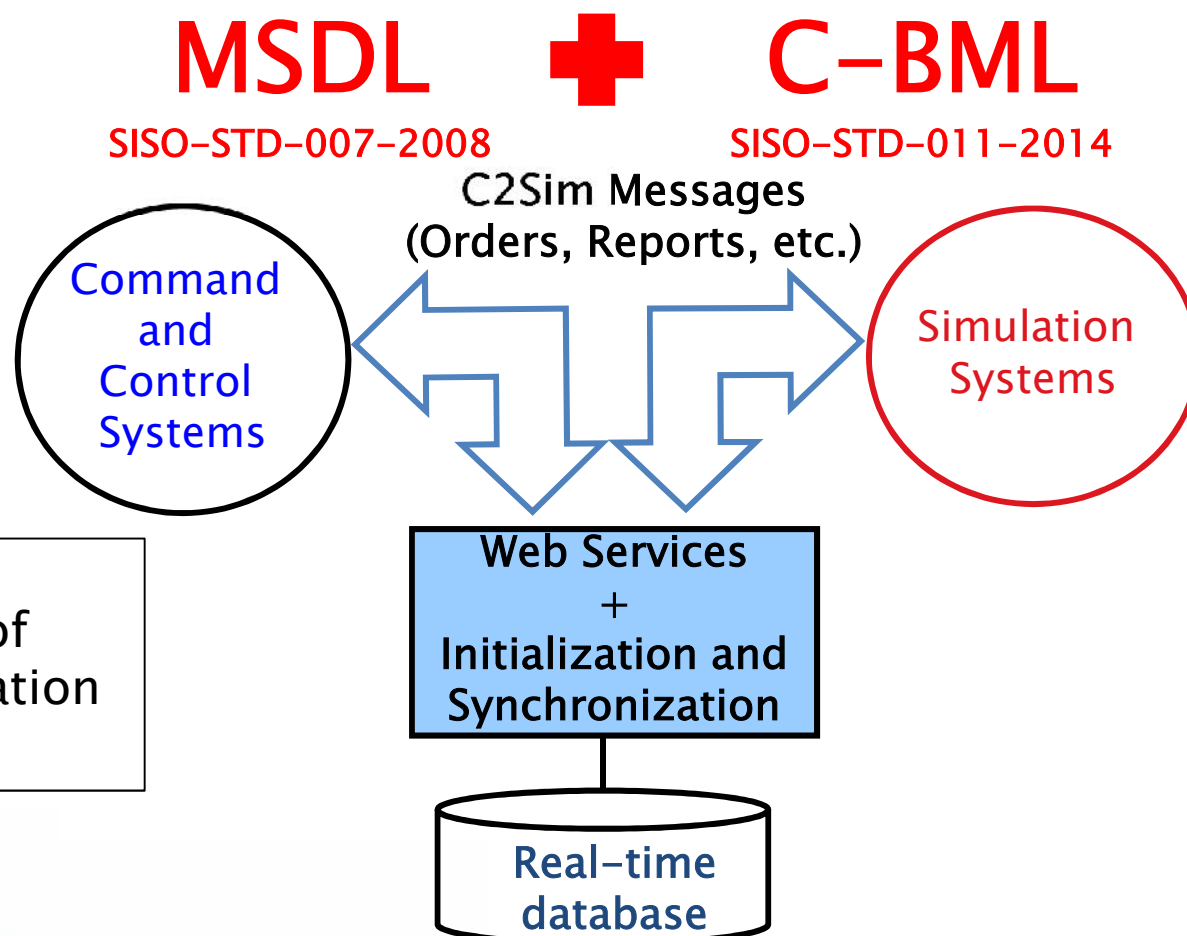


Summary

- ▶ C2Sim Interoperability Language short intro
 - Extension to UAxS
- ▶ C2 of UAxS with C2Sim prodrome C-BML
- ▶ UAxS Simulation Use Case
 - Conceptual Scenario Development Process
- ▶ Requirements for C2Sim extension to C2Sim
- ▶ Distributed Simulation Environment



C2-Simulation Interoperability Language



NMSG-145
“Operationalization of
Standardized C2-Simulation
Interoperability”



C2-Simulation Interoperability Language

Logical Data Model (LDM)

C2Sim



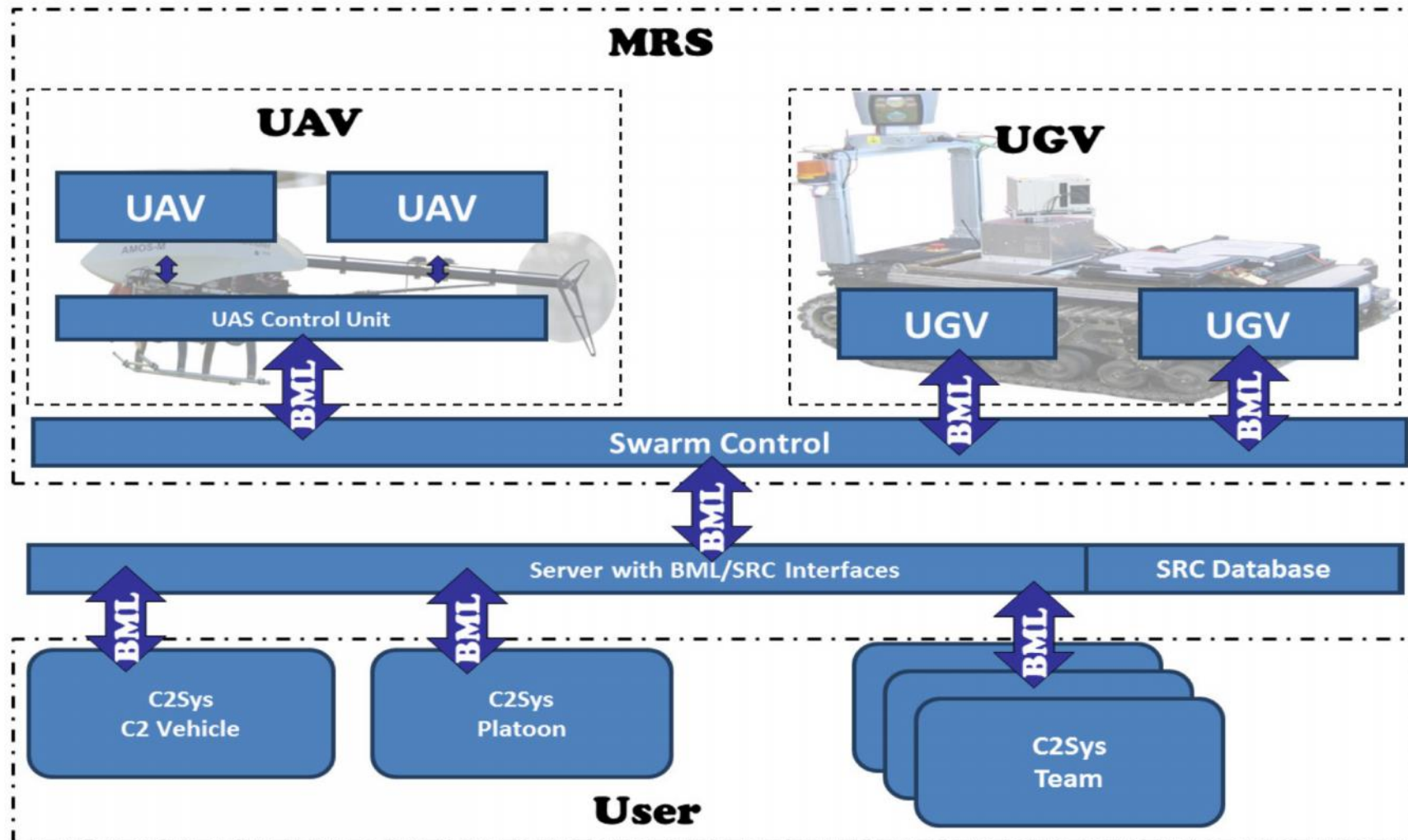
Grammar

Goal

Requirements for C2Sim LDM extension to

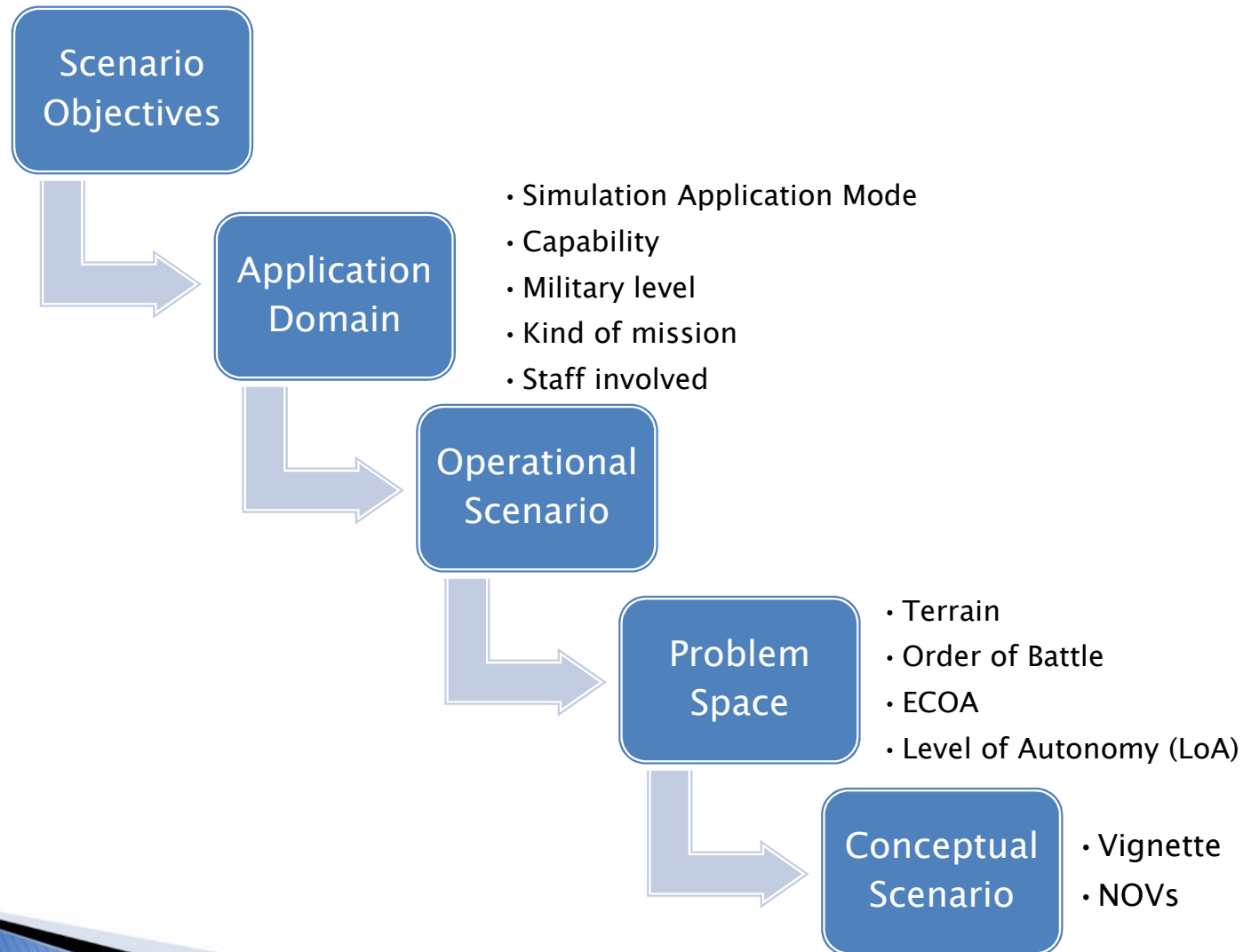
UAXS Application Area

- ▶ Command & Control of Unmanned Autonomous Systems (UAXS)
- ▶ C2 systems & Simulators interoperability

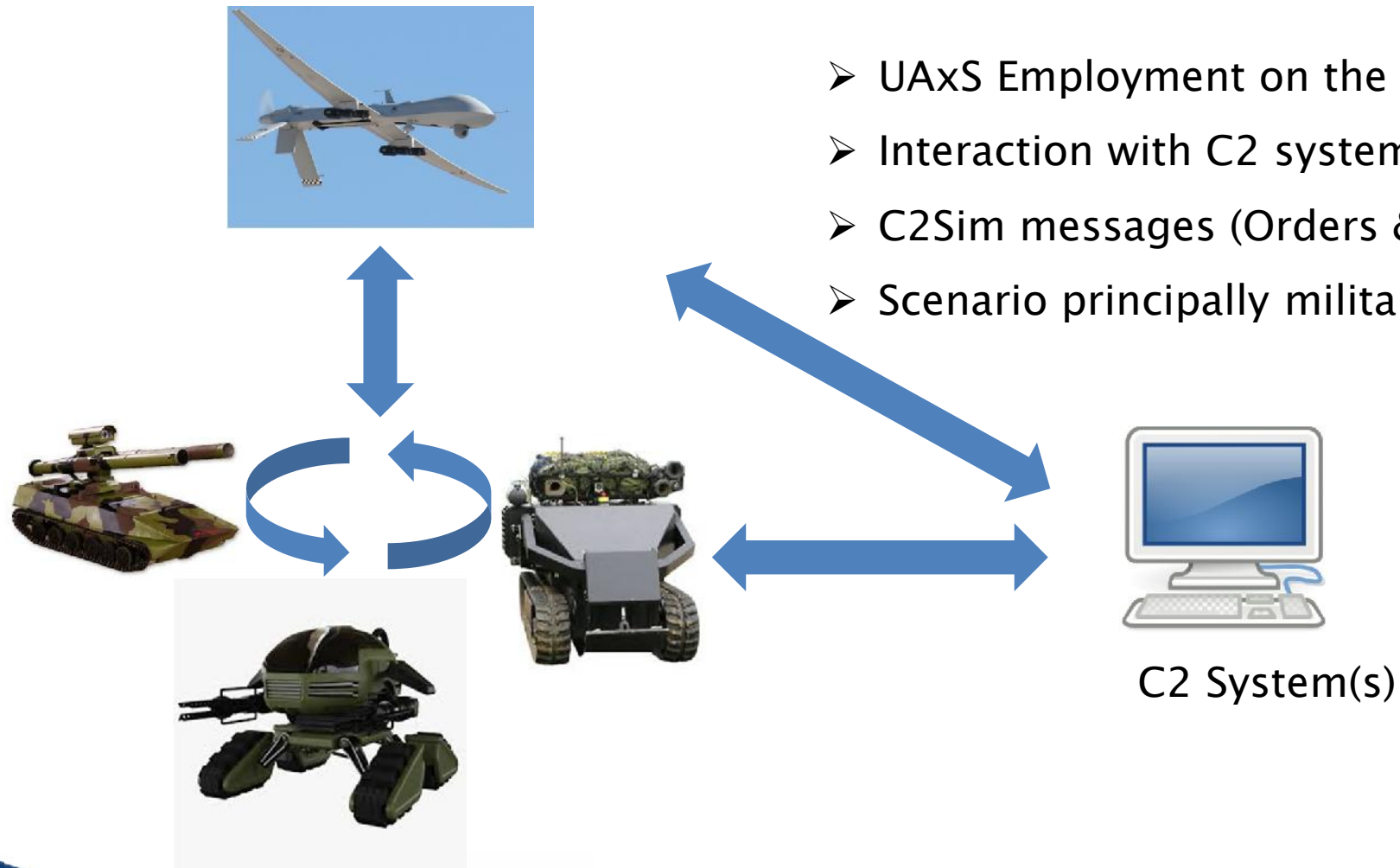




UAXS Simulation SISO GDS Process



UAXS Simulation Objectives





UAXS Simulation Application Domain

1. Simulation Application Mode: CD&E;
2. Capability: Unmanned Autonomous Systems Analysis and Interoperability with C2 systems;
3. Military level: Tactical to Operational;
4. Kind of mission: The defense of an area of interest, in presence of probable CBRN contaminants;
5. Staff involved: Subject Matter Experts (SMEs) on Robotics, Interoperability Languages and Data Models and on Modelling and Simulation (M&S).



UAxS Simulation Operational Scenario

Friendly unit at the coy or platoon level

Threat consist of unknown enemy unit

Terrain without any significant natural obstacles

UAxS in two domains:

1. Air for reconnaissance (find and detect the threat, possibly identify);
2. Land for neutralize the threat.

UAxS perform the assigned mission autonomously according to their level of autonomy.

Message exchange (tasking and reporting) and scenario initialization in C2Sim.

The reports should contain at least the position of the enemy, but also if possible the type and level of the enemy unit and if the area contains any CBRN agents.



UAXS Simulation Problem Space

Terrain

- Open field without significant obstacles (flat or hilly terrain, with or without little woods)

Order of Battle

- Friendly forces should be an infantry company or platoon, as well as the enemy unit, even if unknown at first to the friendly forces
- UAaV for reconnaissance mission, so Mini (from 1 to 3 kg) or Small (from 3 to 10 kg) UAaVs equipped with sensors for Intelligence, Surveillance, Target Acquisition and Reconnaissance (ISTAR) missions and no weapons. Some CBRN sensors should be included.
- UAgV has to be more robust and built for combat mission, especially in contaminated areas, light armored and armed with machine guns and some anti-tank missiles

ECOA

- Depends on the characteristics of the terrain, whose analysis as well as of other related factors is traditionally performed through the Intelligence Preparation of Battlefield (IPB). If a CBRN attack should be included in the scenario has to be decided.

Level of Autonomy (LoA)



UAXS Simulation Level of Autonomy (LoA)

0

- The human only gathers and monitors (defined as filtering, prioritizing and understanding) all data.
- The human analyses all data, predicts and interprets data.
- The system does not assist in or perform ranking tasks. The human must do it all.
- The human alone can execute decision.

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- The system is used for a specific mission.
- The system gathers and displays all the information to the human, but it highlights the non-prioritized, relevant information for the user.
- The system analyses the information to provide data and makes predictions, though the human is responsible for interpretation of the data.
- Both the human and the system perform ranking tasks but the results from the system are considered prime.
- The system allows the human a pre-programmed context-dependent time to veto before execution. The human shadows for contingencies.

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- The system is tasked with a specific mission.
- The system gathers, filters, and prioritizes data.
- The system integrates, interprets data and makes predictions.
- The system performs final ranking.
- The system executes automatically and does not allow any human interaction. Final results are displayed to the human.



UAXS Simulation Conceptual Scenario

Terrain

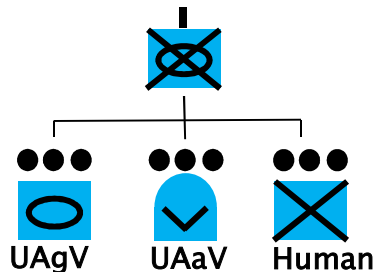
- The mechanized infantry coy is located in a known area, in open countryside environment with little woods. Recent maps are available with roadmap of the area

Order of Battle

• Red Forces



• Blue Forces



EOA

- An unknown enemy unit attacks an isolated house with a bomb with spread locally a chemical warfare substance. The goal is to attire the blue forces and attack them. The enemy unit is located along the route between the attacked house and the blue force position, exploiting every coverage opportunity offered by the terrain in order to ambush them.

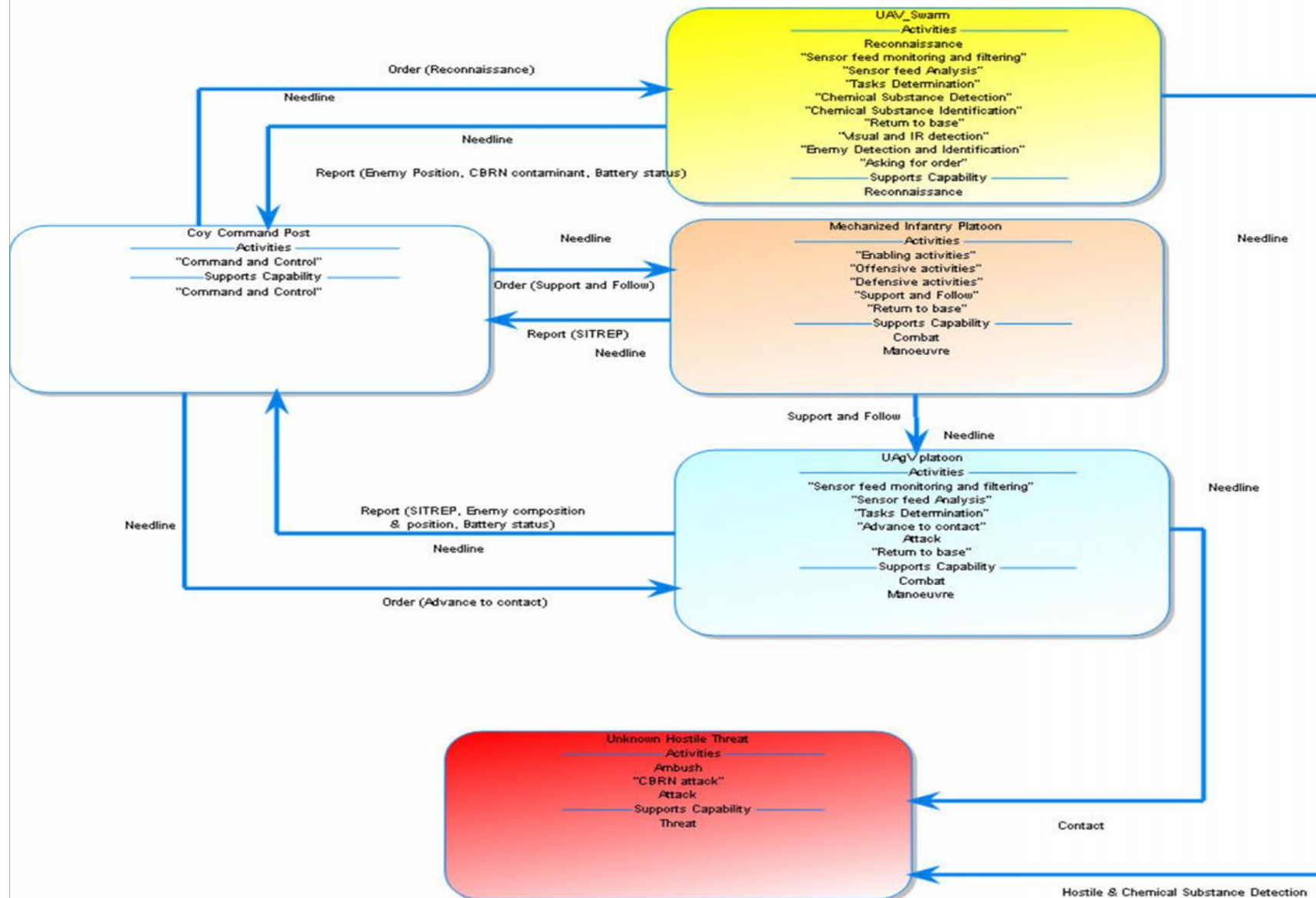
Level of Autonomy (LoA)

- 3: they are tasked with a mission and they can elaborate intermediate tasks, but always ask the humans for confirmation. They always report back or display to humans the feedbacks of their sensors.



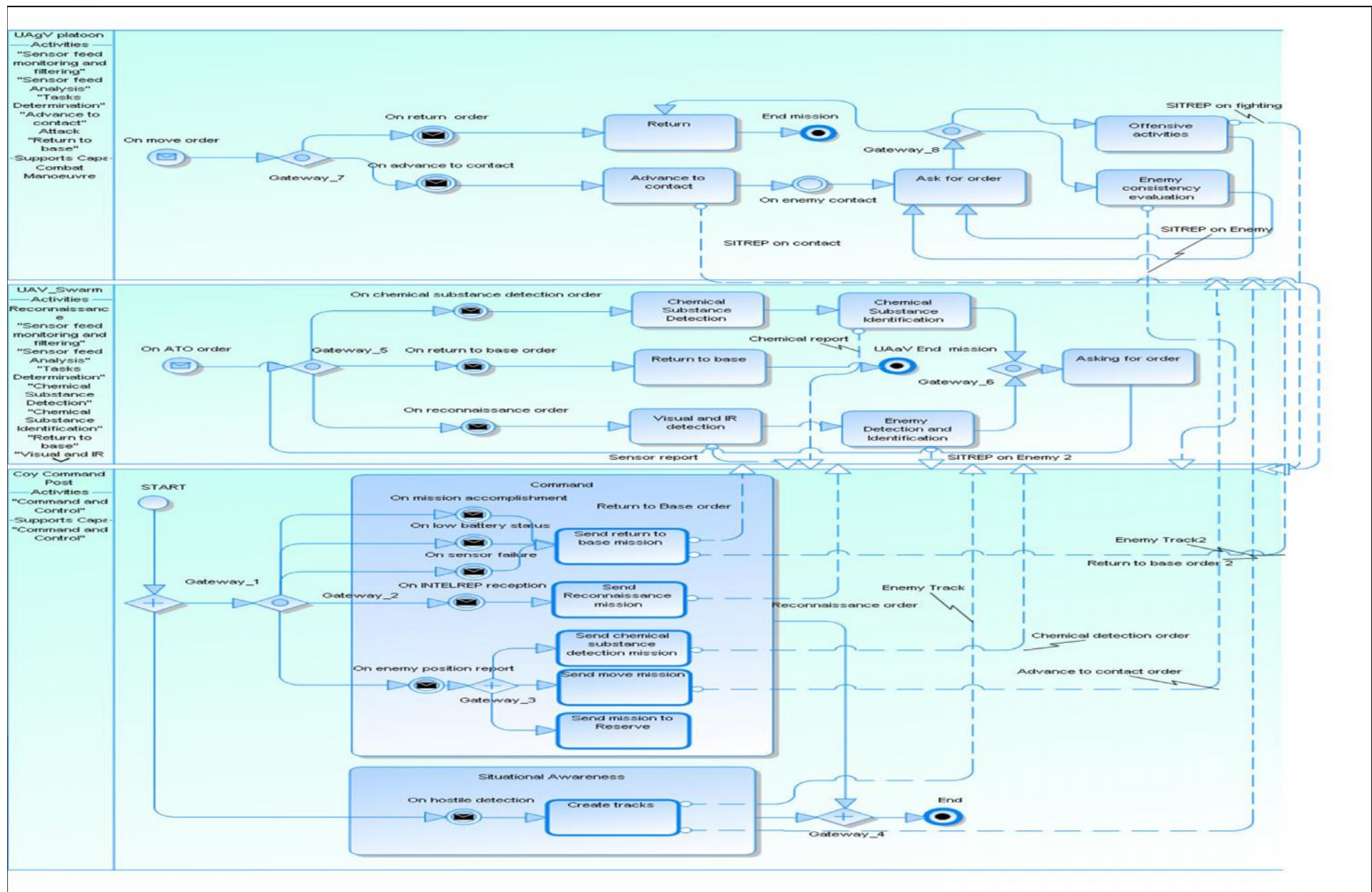
UAXS Conceptual Scenario Vignette

1. The coy CP orders the UAaV platoon leader to reconnoiter the area around the house and to search and acquire enemy targets.
2. UAaV swarm performs autonomously the mission and finds the enemy unit.
3. UAaV swarm reports backs to coy CP about the enemy position and consistency and then asks for authorization to continue the mission towards the attacked house.
4. UAgV platoon is ordered to “Advance to contact” toward the reported enemy position.
5. The human infantry platoon is ordered to be prepared to “Support and Follow” the UAgV platoon.
6. UAaV swarm reports backs about the chemical contaminant as soon as it reaches the contaminated area and then asks for authorization to come back.
7. The coy CP orders the UAaV swarm to come back to the base.
8. UAgV platoon enters in contact with enemy and reports back about confirmed enemy consistency and any possible fighting, then it waits for orders.
9. The coy CP orders the UAgV platoon to come back to the base
10. Both UAaVs and UAgVs return to their own bases.





NAF OV-5





Initialization/Tasking

Requirement	Description
Domain of operation/Type of UAxV	Type of unmanned autonomous platform (if ground, surface, underwater, air platform)
UAxV Size	Micro/Mini/Small/Medium/Large for UAaV Light/Medium/High armored for UAgV Similar classification for maritime platforms.
UAxV specialization	Combat/Reconnaissance/Multipurpose/Target
Level of Autonomy	It fixes the behavior of the UAxS and its interaction with its HQ (rate of reports, coordination and/or confirmation on the tasks to be performed, possibility to make its own decisions based on the feedbacks from the environments).
Level of Interoperability	For aerial platforms (UAaV) STANAG 4586 [24] defines different levels of interoperability with the control station (1–5). It could be a parameter for such kind of platform.
Battery Status	Energy level in the battery of the UAxV at the mission start.
Leadership/Role	Information about the role of the single UAxV inside the swarm or platoon, even if it is not mandatory that the UAxV should have different roles or a leader.



Reporting

Requirement	Description
Battery Status	Energy level remaining in the battery of the UAxV. This information is quite critical for continuing or not the mission.
Suggested Task	Depending on the LoA of the UAxV, it could suggest the following task to be performed according to the external conditions (terrain, enemy behavior, weather, etc.)
CBRN Sensor feedback	This, in the case of the conceptual scenario discussed, contains the information about the chemical substance detected.
Connectivity Status	This is an example of the information about parameters of the internal status of the UAxV (other parameters could be temperature, sensor operation, etc.).
Leadership/Role	Information about the role of the single UAxV inside the swarm or platoon. It could change depending on several factors, even if it is not mandatory that the UAxV should have different roles or a leader.


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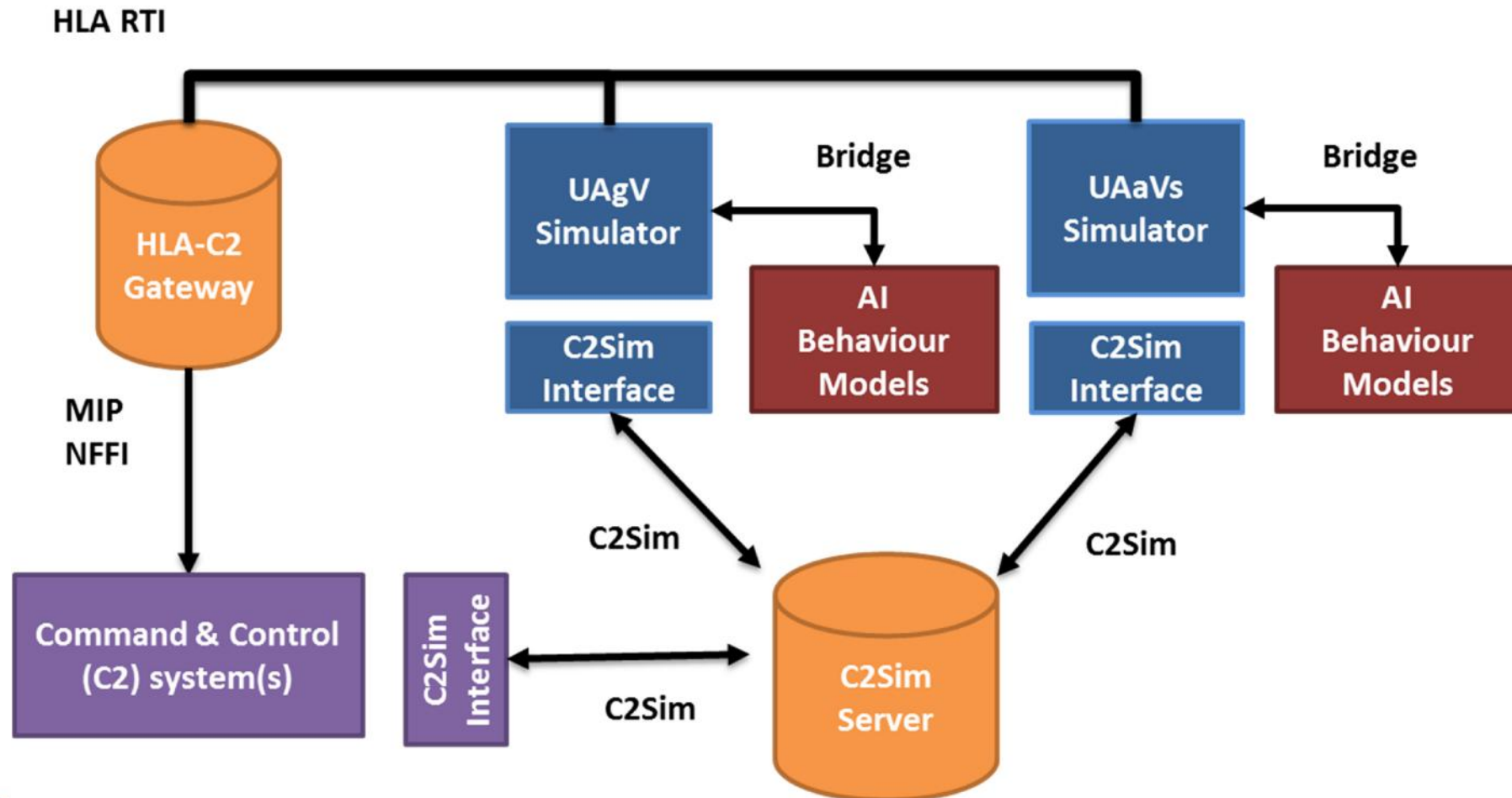
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DISTRIBUTED SIMULATION ENVIRONMENT for UAxS





Q & A

